

PROBLEM SET 5

ECON301—ECONOMETRICS I

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Instructions: Due on 23/04/2022; scan or type your work and submit it through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

Note: in the lecture notes and videos, I usually denote vectors by underlining them, for example $\underline{x}$, and I denote matrices in capital letters. When typesetting it, I use **bold notation and upright letters** to denote vectors and matrices, i.e., $\mathbf{x}$ and $\mathbf{M}$ for example.

1. (a) The null hypothesis is $H_0 : \gamma_1 \leq 0$ versus the alternative $H_1 : \gamma_1 > 0$. This is a right-sided test. The t-statistic for this test is:

$$t_{calc} = \frac{1.49883206}{\sqrt{0.2913855}} = 2.776636$$

The critical value for a right-sided test is $z_{0.05} = 1.644854$. Since the t statistic is larger than the critical value, thus falling in the rejection region, we conclude that we reject the null at a significance level of 5%. The p-value is $\Pr(N(0, 1) > 2.776636) = 0.00274623$.

- (b) The null hypothesis is $H_0 : \gamma_1 \leq \gamma_2$ versus the alternative $H_1 : \gamma_1 > \gamma_2$. This can be restated as $H_0 : \lambda^T \gamma \leq 0$ vs $H_1 : \lambda^T \gamma > 0$, where $\lambda = (0, 1, -1)^T$ and $\gamma = (\gamma_0, \gamma_1, \gamma_2)^T$. This is a right-sided test. The test statistic is

$$t_{calc} = \frac{\lambda^T \hat{\gamma}}{\sqrt{\lambda^T \hat{\Omega} \lambda}} = \frac{1.49883206 - 0.48985174}{0.615347} = 1.639693.$$

The critical value is again $z_{0.05} = 1.644854$. Since the t statistic is smaller than the critical value, we fail to reject the null hypothesis. The p-value of the test is $\Pr(N(0, 1) > 1.639693) = 0.05053448$.

- (c) The null hypothesis is $H_0 : \gamma_2 \leq 1$ versus the alternative $H_1 : \gamma_2 > 1$. This is a right-sided test. The t-statistic for this test is:

$$t_{calc} = \frac{0.4898517 - 1}{\sqrt{0.0104130}} = -4.999292$$

Once again, the critical value is $z_{0.05} = 1.644854$, so we fail to reject the null hypothesis. The p-value of the test is $\Pr(N(0, 1) > -4.999292) = 0.9999997$.

- (d) The null hypothesis is $H_0 : \gamma_2 \leq 1.2$ versus the alternative $H_1 : \gamma_2 > 1.2$. This is a right-sided test. The t-statistic for this test is:

$$t_{calc} = \frac{0.4898517 - 1.2}{\sqrt{0.0104130}} = -6.959228$$

Once again, the critical value is $z_{0.05} = 1.644854$, so we fail to reject the null hypothesis. The p-value of the test is $\Pr(N(0, 1) > -6.959228) \approx 1$.

- (e) The null hypothesis is $H_0 : \gamma_1 + \gamma_2 \leq 1$ versus the alternative $H_1 : \gamma_1 + \gamma_2 > 1$. This can be restated as $H_0 : \lambda^T \gamma \leq 1$ vs $H_1 : \lambda^T \gamma > 1$, where $\lambda = (0, 1, 1)^T$ and $\gamma = (\gamma_0, \gamma_1, \gamma_2)^T$. This is a right-sided test. The test statistic is

$$t_{calc} = \frac{\lambda^T \hat{\gamma}}{\sqrt{\lambda^T \hat{\Omega} \lambda}} = \frac{1.49883206 + 0.48985174 - 1}{0.4742838} = 2.084583.$$

The critical value is again $z_{0.05} = 1.644854$. Since the t statistic is larger than the critical value, we reject the null, therefore, there is evidence of increasing returns to scale at the 5% significance level. The p-value of the test is $\Pr(N(0, 1) > 2.084583) = 0.01855358$.

- (f) The null hypothesis is $H_0 : \mathbf{H}_S \gamma = \mathbf{0}$ versus $H_1 : \mathbf{H}_S \gamma \neq \mathbf{0}$, where

$$\mathbf{H}_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The Wald test statistic is

$$W_{calc} = (\mathbf{H}_S \hat{\gamma})^T (\mathbf{H}_S \hat{\Omega} \mathbf{H}_S^T)^{-1} (\mathbf{H}_S \hat{\gamma}) = 96.13485.$$

We need to compare this to the critical value at the 5% of a chi-square distribution with 2 degrees of freedom. This critical value is $\chi_{0.05}^2(2) = 5.991465$. Since the Wald statistic is larger than the critical value we reject the null at the 5% level. This means that there is evidence that at least one of the output elasticities is different from zero. The p-value of the test is $\Pr(\chi^2(2) > 96.13485) \approx 0$.

2. Note: in this problem I am denoting the population BLP intercept as α and the population BLP slope as β . This problem is using the notation of lectures 2 and 3.

- (a) We have that $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$. By the weak law of large numbers $\bar{Y} \rightarrow_P \mu_Y$, $\bar{X} \rightarrow_P \mu_X$. By theorem 4.3 and the consistency of $\hat{\beta}_1$ we have that $\hat{\beta}_0 \rightarrow_P \mu_Y - \beta \mu_X = \alpha$.

- (b) We have that

$$\frac{\hat{\beta}_1 - \beta}{\hat{\phi}/\sqrt{n}} \sim AN(0, 1).$$

On the other hand, we have that the consistency of $\hat{\phi}^2$ and applying theorem 4.3 twice yields $\hat{\phi}/\phi \rightarrow_P 1$. Therefore, by Slutsky's Theorem (Thm. 4.6) we have that

$$\frac{\frac{\hat{\beta}_1 - \beta}{\hat{\phi}/\sqrt{n}}}{\hat{\phi}/\phi} = \frac{\hat{\beta}_1 - \beta}{\hat{\phi}/\sqrt{n}} \sim AN(0, 1).$$

Therefore, $T = \frac{\hat{\beta}_1 - \beta}{SE(\hat{\beta}_1)}$ where $SE(\hat{\beta}_1) = \hat{\phi}/\sqrt{n}$ is asymptotically standard normal and this is the asymptotic pivot we need to make inference on β .

- (c) In here, I am going to change the notation with respect to the problem set question to try not to confuse you. The asymptotic $(1 - \theta)\%$ confidence interval for β is $[\hat{\beta}_1 \pm z_{\theta/2} SE(\hat{\beta}_1)]$ where $z_{\theta/2}$ is the $\theta/2$ critical value for a standard normal distribution, i.e., $\Pr(N(0, 1) > z_{\theta/2}) = 1 - \theta/2$.
- (d) I am also going to change the notation here a little bit with what was written in the problem set so as to not confuse you. Consider the test statistic: $T_{calc} = \frac{\hat{\beta}_1 - \beta^\circ}{SE(\hat{\beta}_1)}$.

Hypothesis statement:	$H_0 : \beta = \beta^\circ$ vs. $H_1 : \beta \neq \beta^\circ$	$H_0 : \beta \leq \beta^\circ$ vs. $H_1 : \beta > \beta^\circ$
Test statistic and asymptotic distribution under H_0 :	$T_{calc} \sim AN(0, 1)$	$T_{calc} \sim AN(0, 1)$
Critical region for test at $\theta\%$ significance level:	$ T_{calc} > z_{\theta/2}$	$T_{calc} > z_\theta$
P-value	$\Pr(N(0, 1) > T_{calc}) = 2(1 - \Pr(N(0, 1) < T_{calc}))$	$\Pr(N(0, 1) > T_{calc})$

3. Table 1 will be helpful.

Table 1: Calculations for task 3.

Term Z_i	X_i	Y_i	$(X_i - \bar{X})^2$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$\hat{\epsilon}_i$	$((X_i - \bar{X})\hat{\epsilon}_i)^2$
$n = 4$	5.00	8.00	4.00	6.00	-0.40	0.64
	4.00	7.50	1.00	2.50	0.80	0.64
	2.00	2.50	1.00	2.50	-0.80	0.64
	1.00	2.00	4.00	6.00	0.40	0.64
Sum $\sum_i Z_i$	12.00	20.00	10.00	17.00	0.00	2.56

- (a) $\hat{\beta}_1 = S_{xy}/S_{xx} = 17/10 = 1.7$, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 20/4 - (17/10)(12/4) = 5 - 51/10 = -1/10 = -0.1$.
- (b) The residuals are in the column titled $\hat{\epsilon}_i$ in table 1 above.
- (c) $\hat{\phi}^2 = n(S_{xx})^{-1} \sum_i ((X_i - \bar{X})\hat{\epsilon}_i)^2 (S_{xx})^{-1} = 4(1/10)(2.56)(1/10) = 0.1024$.
- (d) The standard error of $\hat{\beta}_1$ is $SE(\hat{\beta}_1) = \sqrt{\hat{\phi}^2/n} = \hat{\phi}/\sqrt{n} = \sqrt{0.1024/4} = 0.16$. Therefore, a CI interval at the 95% level is $[1.7 \pm 1.959964(0.16)] = [1.7 \pm 0.3135942] = [1.386406, 2.013594]$.
- (e) The null is $H_0 : \beta = 0$ versus the alternative $H_1 : \beta \neq 0$. The t-statistic for this test is $t_{calc} = 1.7/0.16 = 10.625$. The critical value for a two-sided test from the normal distribution at the 1% level is $z_{0.01/2} = 2.575829$. Therefore, it is clear that we will reject the null hypothesis, so there is evidence at the 1% level that $\beta \neq 0$.

4. (a) The null hypothesis is $H_0 : \mathbf{H}_S \boldsymbol{\beta} = \mathbf{0}$ versus $H_1 : \mathbf{H}_S \boldsymbol{\beta} \neq \mathbf{0}$, where

$$\mathbf{H}_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The Wald test statistic is

$$W_{calc} = (\mathbf{H}_S \hat{\boldsymbol{\beta}})^\top (\mathbf{H}_S \hat{\boldsymbol{\Sigma}} \mathbf{H}_S^\top)^{-1} (\mathbf{H}_S \hat{\boldsymbol{\beta}}) = 306.956.$$

We need to compare this to the critical value at the 5% of a chi-square distribution with 2 degrees of freedom. This critical value is $\chi^2_{0.05}(2) = 5.991465$. Since the Wald statistic is larger than the critical value we reject the null at the 5% level. This means that there is evidence that at least one of the coefficients is different from zero. The p-value of the test is $\Pr(\chi^2(2) > 306.956) \approx 0$.

- (b) The standard error is $SE(\hat{\beta}_1) = \sqrt{0.004506} = 0.06712675$. The critical value for a confidence interval at the 95% level is $z_{0.05/2} = 1.959964$. Therefore, the confidence interval at the 95% level is $[-1.003587 \pm 1.959964(0.06712675)] = [-1.003587 \pm 0.131566] = [-1.135153, -0.872021]$. As we can see, the confidence interval includes only negative numbers. The interpretation is that 1 more point in the average number of students per teacher decreases the *ET* score by 1.14 and 0.87 points (in percentage points since the variable is in percent). This is consistent with common sense: the more students there are per teacher, the worse the educational achievement will be.

- (c) The null hypothesis is $H_0 : \beta_2 \leq 0$ versus the alternative $H_1 : \beta_2 > 0$. The t-statistic is

$$t_{calc} = 7.332466 / \sqrt{7.328432} = 2.708598.$$

Once again, this is a right-sided test, so the critical value is $z_{0.05} = 1.644854$. Clearly, we reject the null at the 5% significance level. The p-value is $\Pr(N(0, 1) > 2.708598) = 0.003378409$.

5. For this problem, I will fully use matrix notation. I also compute some other matrices that you do not need to compute by yourselves but which I compute here for two reasons: 1) They are interesting on their own, 2) This is a small and feasible example in which it is easy to do.

- (a) The regressor matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix},$$

and the vector of dependent variables is

$$\mathbf{Y} = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 2 \\ 3 \end{bmatrix}.$$

Then we have

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 5 & 3 \\ 3 & 3 \end{bmatrix}, \quad (\mathbf{X}^\top \mathbf{X})^{-1} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 5/6 \end{bmatrix}, \quad (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \begin{bmatrix} 0 & 0 & 0 & 1/2 & 1/2 \\ 1/3 & 1/3 & 1/3 & -1/2 & -1/2 \end{bmatrix},$$

and finally

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \begin{bmatrix} 5/2 \\ -5/6 \end{bmatrix}.$$

You do not need to compute this, but is nevertheless interesting to do it: the projection matrix $\mathbf{P}_X$ is

$$\mathbf{P}_X = \begin{bmatrix} 1/3 & 1/3 & 1/3 & 0 & 0 \\ 1/3 & 1/3 & 1/3 & 0 & 0 \\ 1/3 & 1/3 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 0 & 1/2 & 1/2 \end{bmatrix}$$

The vector of fitted values is

$$\hat{\mathbf{Y}} = \mathbf{P}_X \mathbf{Y} = \begin{bmatrix} 5/3 \\ 5/3 \\ 5/3 \\ 5/2 \\ 5/2 \end{bmatrix}$$

The residual making projection matrix $\mathbf{N}_X$ is

$$\mathbf{N}_X = \begin{bmatrix} 2/3 & -1/3 & -1/3 & 0 & 0 \\ -1/3 & 2/3 & -1/3 & 0 & 0 \\ -1/3 & -1/3 & 2/3 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & -1/2 \\ 0 & 0 & 0 & -1/2 & 1/2 \end{bmatrix}.$$

Therefore, the vector of residuals is

$$\hat{\epsilon} = \begin{bmatrix} 1/3 \\ 4/3 \\ -5/3 \\ -1/2 \\ 1/2 \end{bmatrix}.$$

To compute the $\hat{\Omega}$ matrix of variance-covariance estimates, we need a new matrix, the $\hat{\mathbf{D}}$ which has the squared residuals in the diagonal:

$$\hat{\mathbf{D}} = \text{diag}(\hat{\epsilon}_i^2) = \begin{bmatrix} 1/9 & 0 & 0 & 0 & 0 \\ 0 & 16/9 & 0 & 0 & 0 \\ 0 & 0 & 25/9 & 0 & 0 \\ 0 & 0 & 0 & 1/4 & 0 \\ 0 & 0 & 0 & 0 & 1/4 \end{bmatrix}.$$

Then, the “sandwiched” matrix in the definition of $\hat{\Omega}$ is

$$\mathbf{X}^\top \hat{\mathbf{D}} \mathbf{X} = \begin{bmatrix} 31/6 & 14/3 \\ 14/3 & 14/3 \end{bmatrix}.$$

Finally, we get

$$\hat{\Omega} = (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \hat{\mathbf{D}} \mathbf{X}) (\mathbf{X}^\top \mathbf{X})^{-1} = \begin{bmatrix} 1/8 & -1/8 \\ -1/8 & 139/216 \end{bmatrix}$$

(b) See the attached R script called `problemset5-auxiliary-routines.R`.

6. (a) Note that $\epsilon_i = Y_i - \beta_0 - \beta_1 X_i$, where β_0 and β_1 are the true population BLP parameters. Then W_i is only a function of (X_i, Y_i) , i.e., $W_i = h(X_i, Y_i)$ where $h(X_i, Y_i) = (X_i - \mu_X)\epsilon_i$. By assumption 1M, we have that $h(X_i, Y_i)$ is independent of $h(X_j, Y_j)$ for all $i \neq j$. Therefore, W_i is independent of W_j .
- (b) Theorems 2.7 and 2.8 together with assumption 1M imply that $E(\epsilon_i) = 0$ and that $E(X_i \epsilon_i) = 0$. Therefore $E(W_i) = E(X_i \epsilon_i - \mu_X \epsilon_i) = E(X_i \epsilon_i) - \mu_X E(\epsilon_i) = 0$.
- (c) Since $E(W_i) = 0$ then

$$\text{Var}(W_i) = E(W_i^2) = E((X_i - \mu_X)^2 \epsilon_i^2) = E(X_i^2 \epsilon_i^2 - 2\mu_X X_i \epsilon_i^2 + \mu_X^2 \epsilon_i^2) = E(X_i^2 \epsilon_i^2) - 2\mu_X E(X_i \epsilon_i^2) + \mu_X^2 E(\epsilon_i^2).$$

(d) Since $E(\epsilon_i) = 0$ we have that

$$E(\epsilon_i^2) = \text{Var}(\epsilon_i) = \text{Var}(Y_i - \beta_0 - \beta_1 X_i) = \text{Var}(Y_i) + \beta_1^2 \text{Var}(X_i) - 2\beta_1 \text{Cov}(X_i, Y_i).$$

(e) First note that $\bar{\epsilon} = \frac{1}{n} \sum_i \hat{\epsilon}_i = \frac{1}{n} \sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = \bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X}$. Now, using the hint:

$$\begin{aligned} \frac{1}{n} \sum_i \hat{\epsilon}_i^2 &= \frac{1}{n} \sum_i (\hat{\epsilon}_i - \bar{\epsilon})^2 = \frac{1}{n} \sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i - \bar{Y} + \hat{\beta}_0 + \hat{\beta}_1 \bar{X})^2 = \frac{1}{n} \sum_i [(Y_i - \bar{Y}) - \hat{\beta}_1 (X_i - \bar{X})]^2 \\ &= \frac{1}{n} \sum_i [(Y_i - \bar{Y})^2 - 2\hat{\beta}_1 (Y_i - \bar{Y})(X_i - \bar{X}) + \hat{\beta}_1^2 (X_i - \bar{X})^2] \\ &= \frac{1}{n} \sum_i (Y_i - \bar{Y})^2 - 2\hat{\beta}_1 \left(\frac{1}{n} \sum_i (Y_i - \bar{Y})(X_i - \bar{X}) \right) + \hat{\beta}_1^2 \left(\frac{1}{n} \sum_i (X_i - \bar{X})^2 \right). \end{aligned} \quad (1)$$

Now, note that we still have that the sequences $\{(Y_i - \bar{Y})^2\}_i$ and $\{(X_i - \bar{X})^2\}_i$ are not i.i.d., so we cannot apply the WLLN directly, but using the techniques from lecture 4 (check slide 6 from lecture 4) we can show that:

$$\begin{aligned} \frac{1}{n} \sum_i (Y_i - \bar{Y})^2 &\rightarrow_P \text{Var}(Y_i) \\ \frac{1}{n} \sum_i (X_i - \bar{X})^2 &\rightarrow_P \text{Var}(X_i) \end{aligned}$$

On the other hand, we also have that the sequence $\{(Y_i - \bar{Y})(X_i - \bar{X})\}_i$ is not i.i.d., but we can do the following:

$$\frac{1}{n} \sum_i (Y_i - \bar{Y})(X_i - \bar{X}) = \frac{1}{n} \sum_i X_i Y_i - \bar{X} \bar{Y}$$

The term $\frac{1}{n} \sum_i X_i Y_i$ obeys the assumptions of the WLLN (theorem 4.4) so that it converges in probability to $E(X_i Y_i)$. On the other hand the WLLN implies that $\bar{X} \rightarrow_P \mu_X$ and $\bar{Y} \rightarrow_P \mu_Y$ so that by theorem 4.3, $\bar{X} \bar{Y} \rightarrow_P \mu_X \mu_Y$. Another application of theorem 4.3 yields that:

$$\frac{1}{n} \sum_i (Y_i - \bar{Y})(X_i - \bar{X}) = \frac{1}{n} \sum_i X_i Y_i - \bar{X} \bar{Y} \rightarrow_P E(X_i Y_i) - \mu_X \mu_Y = \text{Cov}(X_i, Y_i)$$

Assumptions 1M and 2M guarantee that $\hat{\beta}_1 \rightarrow_P \beta_1$ and an application of theorem 4.3 yields that $\hat{\beta}_1^2 \rightarrow_P \beta_1^2$. Combining all of the above together in (1) with yet another one application of theorem 4.3 yields that

$$\frac{1}{n} \sum_i \hat{\epsilon}_i^2 \rightarrow_P \text{Var}(Y_i) - 2\beta_1 \text{Cov}(X_i, Y_i) + \beta_1^2 \text{Var}(X_i) = E(\epsilon_i^2).$$

- (f) We propose the estimator $H_n = \bar{X}^2 \left(\frac{1}{n} \sum_i \hat{\epsilon}_i^2 \right)$. Once again, by theorem 4.3, $\bar{X}^2 \rightarrow_P \mu_X^2$ and the previous task shows the convergence in probability of $\frac{1}{n} \sum_i \hat{\epsilon}_i^2$ to $E(\epsilon_i^2)$. Another application of theorem 4.3 yields that $\bar{X}^2 \left(\frac{1}{n} \sum_i \hat{\epsilon}_i^2 \right) \rightarrow_P \mu_X^2 E(\epsilon_i^2)$.
- (g) We propose the estimator $J_n = \frac{1}{n} \sum_i X_i^2 \hat{\epsilon}_i^2$, which converges in probability to $E(X_i^2 \epsilon_i^2)$. A similar technique as in task (e) can be used to show that it is consistent. The proof is tedious so that is why I am not asking for you to show it. It also requires that we assume the existence of some higher order moments of X_i and Y_i .
- (h) We propose the estimator $L_n = 2\bar{X} \left(\frac{1}{n} \sum_i X_i \hat{\epsilon}_i^2 \right)$, which converges in probability to $2\mu_X E(X_i \epsilon_i^2)$. Once again, a similar technique as in task (e) can be used to show that it is consistent. The proof is also cumbersome so that is why I am not asking you to show it.
- (i) Firstly, note that

$$\begin{aligned} J_n - L_n + H_n &= \frac{1}{n} \sum_i X_i^2 \hat{\epsilon}_i^2 - 2\bar{X} \left(\frac{1}{n} \sum_i X_i \hat{\epsilon}_i^2 \right) + \bar{X}^2 \left(\frac{1}{n} \sum_i \hat{\epsilon}_i^2 \right) \\ &= \frac{1}{n} \sum_i \left(X_i^2 \hat{\epsilon}_i^2 - 2\bar{X} X_i \hat{\epsilon}_i^2 + \bar{X}^2 \hat{\epsilon}_i^2 \right) \\ &= \frac{1}{n} \sum_i [(X_i - \bar{X}) \hat{\epsilon}_i]^2 \\ &= \hat{\sigma}_W^2. \end{aligned}$$

Now, from (f), (g) and (h) and theorem 4.3:

$$\hat{\sigma}_W^2 = J_n - L_n + H_n \rightarrow_P E(X_i^2 \epsilon_i^2) - 2\mu_X E(X_i \epsilon_i^2) + \mu_X^2 E(\epsilon_i^2) = \text{Var}(W_i)$$